First Cross- and Inter-Band Calibrations of the Hyperspectral Imager Suite Using Off-Nadir Quasi-Simultaneous Overpass Counterparts

Hiroki Mizuochi, Satoshi Tsuchida, Satoru Yamamoto, Minoru Urai, Moe Matsuoka, Ayame Ikeda, and Koki Iwao

***Abstract*— The hyperspectral imager suite (HISUI) is a recently commissioned spaceborne hyperspectral sensor that is expected to provide higher-quality information for terrestrial and aquatic monitoring at a global scale. To maximize the utility of the detailed spectra observed by HISUI, calibration and validation activities have been intensively conducted for the past several years. However, research on long-term monitoring of sensor radiometric characteristics and checking inter-band radiometric consistency is still limited. This study provided the first results of cross- and inter-band calibrations using five scenes acquired over a desert calibration site during 2021–2023, and calculated relative cross-calibration coefficients (RCCCs) to calibrate all the HISUI bands. To make full use of the limited amount of data archived over the calibration site, off-nadir scenes obtained by wide-swath counterpart sensors [i.e., the moderate resolution imaging spectroradiometer (MODIS)] were used for the cross-calibration, with correction by a bidirectional reflectance distribution function. The obtained RCCCs successfully mitigated the inter-band radiometric variability of HISUI data, which was consistent with the in situ surface reflectance data observed by the radiometric calibration network (RadCalNet). Time-series analysis further revealed a notable discrepancy between the visible to near-infrared and short-wave infrared (SWIR) spectral domains after the International Space Station (ISS) maneuvers in spring 2022. This article is also a prompt report of this phenomenon and will soon contribute to HISUI radiometric calibration, particularly after the maneuver events.**

***Index Terms*— Bidirectional reflection function (BRDF) correction, cross-calibration, hyperspectral imager suite (HISUI), inter-band calibration, moderate resolution imaging spectroradiometer (MODIS).**

I. Introduction

HYPERSPECTRAL remote sensing has the potential to provide higher-quality terrestrial and aquatic monitoring data due to its ability to observe detailed spectral features of absorption and reflection on the Earth's surface. Recently, more than 20 hyperspectral imagers have been deployed into space [1], most of which are Earth observation missions.

The hyperspectral imager suite (HISUI) is one of the newer spaceborne hyperspectral imagers, developed by Japan's Ministry of Economy Trade and Industry [2], [3], [4]. It was launched and installed on the International Space Station (ISS) in late 2019 and had acquired approximately 500 000 scene datasets through August 2023. The push-broom-type sensor of HISUI observes the approximately 400- to 2500-nm spectral range with 185 bands. Similar to other hyperspectral satellite missions, HISUI is also expected to have wide-ranging applicability to vegetation, agriculture, coastal bathymetry, forestry, oil and metal resources, and Earth sciences [5], [6].

To ensure the reliability of HISUI, calibration activities, including preflight calibration [3], initial check and onboard-lamp calibration after launch [4], and spectral calibration [6], [7], have been performed thoroughly. The remaining tasks are long-term monitoring of sensor radiometric characteristics and checking inter-band radiometric consistency. The latter task is particularly important for the extraction of narrow-band spectral features.

Cross-calibration is useful for accomplishing the remaining tasks. In this approach, the sensor of interest is compared with another orbiting well-calibrated sensor (i.e., reference counterpart) that observes the same ground target with a small overpass time gap. Well-calibrated publicly available satellite data have been popularly used as references, such as the Landsat series, Sentinel-2 Multispectral Instrument (MSI), and moderate resolution imaging spectroradiometer (MODIS) [8], [9], [10], [11]. This provides an advantageous method for long-term calibration and for the calibration of sensors that have no onboard calibration system. Even for sensors with onboard calibration systems, cross-calibration can provide independent information that can mitigate bias from optical path inconsistency between onboard lamp detection and ground target detection.

The challenging issue in cross-calibration for HISUI is a narrow window for obtaining simultaneous nadir overpass (SNO) counterparts [12], [13], [14] over ideal calibration targets (i.e., homogeneous and pseudo-invariant sites [15], [16]), due to the irregular orbit of the ISS. To make full use of the limited number of narrow-swath hyperspectral data acquired over suitable calibration sites, we previously

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developed a combined approach of cross- and inter-band calibrations using Earth-Observing 1 (EO-1) Hyperion observations as test data, and wide-swath multispectral images (from MODIS) as reference data [17]. In our previous study, 18 quasi-SNO pairs [18] were selected over a well-known calibration site [8], [19], [20] from the rich historical archives of Hyperion and MODIS. Using SNO or quasi-SNO pairs can minimize the uncertainty arising from different overpass times and different sensor geometries. Differences in the spectral response function (SRF) between hyperspectral and multispectral sensors, and differences in spatial resolution between narrow-swath and wide-swath sensors, are treated by surface-reflectance-based band adjustments [21], [22] and by considering the spatial sensitivity of the MODIS instantaneous field of view (IFOV) [23], respectively.

The previous approach is beneficial for increasing calibration capabilities for narrow-swath hyperspectral data and is promising for HISUI cross-calibration. However, a further difficulty is that rich data archives of HISUI over ideal calibration sites are not yet available, making it difficult to search for nadir overpass counterparts. Using off-nadir counterparts may cause calibration biases due to the different sensor geometries [9], [10], [11], [18], [24], and may require further refinement of our previous cross- and inter-band calibration approach.

Therefore, as the first attempt at cross- and inter-band calibration for HISUI, the current study aimed to 1) investigate the time-series radiometric characteristics and inter-band radiometric consistency of HISUI data to provide improved calibration parameters and 2) establish an operational algorithm for cross- and inter-band calibration using off-nadir quasi-simultaneous overpass counterparts with consideration of a bidirectional reflection function (BRDF).

The remainder of this article consists of a description of the study site and related satellite data (Section II), and a presentation of the detailed methodology in the cross- and inter-band calibrations to mitigate discrepancies between HISUI and the counterpart sensors (Section III). Then, we demonstrate the cross-calibration results for corresponding bands between sensors and inter-band calibration results for all the HISUI bands (Section IV) and determine the final relative cross-calibration coefficients (RCCCs) to correct HISUI radiance and their potential uncertainty (Section V). Finally, we provide a summary and conclusions (Section VI).

II. Study Site and Data

A. Study Site

As our study site, we selected Railroad Valley Playa (RRVP), a well-known homogeneous desert site for vicarious and cross-calibrations, which is designated as a radiometric calibration network (RadCalNet) site by the Committee on Earth Observation Satellites [16]. The official extent of the site is 1 × 1 km around a central location (38.497°N, 115.690°W, 1435 m asl) [25] (Fig. 1). The site has stand-alone instruments measuring surface reflectance spectra and atmospheric parameters [26], which can be used as validation data for our cross- and inter-band calibration results.

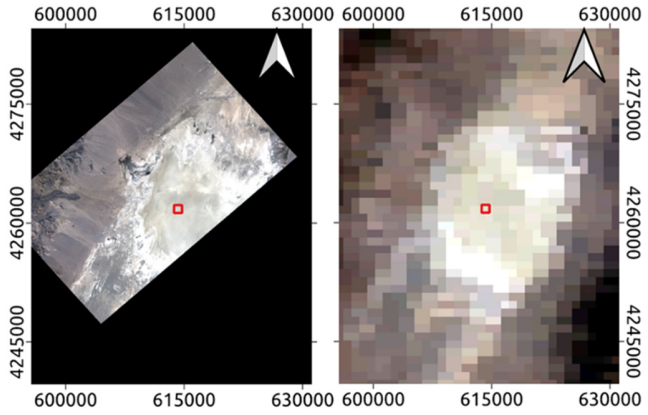

Fig. 1. True-color images by (left) HISUI and (right) MODIS onboard Aqua, acquired on 2021/10/01 over the RRVP site. The images are shown on a UTM zone 11 projection with a meter scale. The red squares show the study area, centered at the RRVP RadCalNet site. The HISUI color composite was red: band 26 (655 nm); green: band 16 (555 nm); and blue: band 10 (495 nm).

Our research team has repeatedly conducted field campaigns around the RRVP site from 2000 to 2017 using a FieldSpec spectroradiometer (Malvern Panalytical Ltd., U.K.) and a Spectralon diffuse reflectance standard panel (Labsphere Inc., USA) to obtain the spectral profile of the ground surface under various conditions [20]. The 11 obtained ground spectral profiles can be used to correct differences in the sensor spectral response (see Section III-A).

B. Satellite Data

Raw HISUI data transported from ISS by spaceship were processed on the calibration data archive system (CDAS) at the Geological Survey of Japan. In the process, radiometric calibration parameters determined by previous calibrations [27] were applied to the raw digital number data to create radiance images (Level 1R) in units of W/(m^2 sr μm). In this process, cross-track response non-uniformity (RNU) was also corrected based on the previous calibration activity over homogeneous desert scenes. The Level 1R images were then geometrically corrected and projected on a universal transverse mercator (UTM) zone 11 projection with the WGS84 datum (Level 1G), and these were used in the current study.

Corresponding to the two different array detectors, image data in the visible to near infrared (VNIR) and short-wave infrared (SWIR) spectral domains were separately provided. For the VNIR domain covering 400–980 nm with 10-nm intervals, 58 band images were available, whereas for the SWIR domain covering 895–2481 nm with 12.5-nm intervals, 127 band images were available. Further specifications and processing of the HISUI products in the CDAS can be found in [6] and [7].

Cloud-free HISUI images over the RRVP study site were identified, and five scenes acquired on 2021/08/02, 2021/10/01, 2022/02/19, 2022/08/03, and 2023/06/23 were selected. Using Sentinel-2 MSI images on 2022/02/19 as a geolocation reference for phase-only correlation [28] corrected a geolocation error in HISUI Level 1G (i.e., co-registration). This treatment was mandated to correct a Level 1G geolocation

TABLE I
SPECIFICATIONS OF THE HISUI AND MODIS DATA OVER THE STUDY SITE ON EACH COLLECTION DATE (YYYY/MM/DD)

Parameter	2021/08/02	2021/10/01	2022/02/19	2022/08/03	2023/06/23
HISUI overpass time[a] and solar zenith angle	20:57:53 (local time 13:57:53*), 25.76°	21:15:18 (local time 14:15:18*), 48.30°	20:19:53 (local time 12:19:53), 49.81°	20:10:21 (local time 13:10:31*), 21.69°	18:44:22 (local time 11:44:22*), 19.76°
HISUI scene cloud rate[b]	7%	7%	6%	2%	0%
Terra/MODIS overpass time and solar zenith angle[c]	19:10:00 (local time 12:10:00*), 22.47°	17:55:00 (local time 10:55:00*), 47.35°	19:00:00 (local time 11:00:00), 51.22°	(A) 17:35:00 (local time 10:35:00*), 35.78° (B) 19:10:00 (local time 12:10:00*), 22.46°	(A) 17:20:00 (local time 10:20:00*), 33.87° (B) 18:55:00 (local time 11:55:00*), 17.84°
Terra/MODIS sensor zenith angle[c]	52.13°	54.08°	46.46°	(A) 62.96° (B) 61.47°	(A) 64.52° (B) 60.22°
Aqua/MODIS overpass time and solar zenith angle	20:50:00 (local time 13:50:00*), 25.37°	21:15:00 (local time 14:15:00*), 48.71°	20:45:00 (local time 12:45:00), 50.99°	20:40:00 (local time 13:40:00*), 24.36°	20:40:00 (local time 13:40:00*), 19.21°
Aqua/MODIS sensor zenith angle	4.43°	43.54°	9.13°	21.57°	41.30°
RadCalNet observation time[d]	-	21:30:00 (local time 14:30:00*)	20:30:00 (local time 12:30:00)	20:00:00 (local time 13:00:00*)	18:30:00 (local time 11:30:00*)

[a]Approximate overpass time is shown in UTC and local time (UTC-8 in standard time, or UTC-7 in summer time).
[b]Percentage of cloud cover over the entire HISUI scene. For the study site (1 km×1 km), we confirmed that there were no cloud pixels by visual inspection and the cloud flag provided with the HISUI products.
[c](A) and (B) on 2022/08/03 denote the two MODIS/Terra observations at 17:35 and 19:10 UTC, respectively, and those on 2023/06/23 denote them at 17:20 and 18:55 UTC, respectively.
[d]RadCalNet data were used for validation of the calibration result, except for 2021/08/02 due to data unavailability.
*Local time is displayed in summer time (UTC-7).

error (parallel displacement), which was sometimes over several to ten kilometers at maxima, which would obviously affect cross-comparison with MODIS.

MODIS is one of the most popular free satellite data sources [29] that offers well-calibrated long-term data archives. Due to its wide swath (2330 km) and constellation by the Terra and Aqua satellites, daily observations at 250- to 1000-m spatial resolution are available at the global scale. A variety of multilevel products allow the convenient implementation of various remote sensing algorithms and applications.

MODIS has seven terrestrial bands, centered at 645 nm (red), 859 nm (NIR), 469 nm (blue), 555 nm (green), 1240 nm, 1640 nm, and 2130 nm (SWIR). For MODIS onboard Aqua, band 6 (1640 nm) cannot be used due to a detector failure [30]; thus, seven bands of Terra/MODIS and six bands of Aqua/MODIS were used for our cross-calibration purpose.

Daily calibrated radiance products from both Terra (MOD021KM) and Aqua (MYD021KM), and corresponding geolocation information (MOD03 and MYD03, respectively) were downloaded from the Earthdata Search website [31] on the same observation dates as those of HISUI. The original hdf format data on the sinusoidal projection were converted to the GeoTIFF format on the UTM zone 11 projection for consistency with HISUI data, using the HDF-EOS to GeoTIFF conversion tool. In this process, radiance images for each band and sensor geometry (solar and sensor zenith and azimuthal angle) were extracted. In addition, a 16-day composite product that provides BRDF parameters (MCD43A1) [32] was obtained from the same website for the implementation of BRDF corrections.

Table I compares the HISUI and MODIS pairs used in the current study, showing overpass times and sensor zenith angles. The sensor zenith angle of HISUI can be assumed to be zero (i.e., approximately a nadir observation) because it has no pointing scheme. HISUI tended to pass over the RRVP site during 20:00–21:00 UTC (12:00–13:00 local time), which is more consistent with the Aqua/MODIS overpass time, but some of the Terra/MODIS data acquired around 19:00 UTC were also likely to make quasi-simultaneous cross-calibration pairs. The Terra/MODIS overpass around 17:00–18:00 UTC often has a 2- to 3-h difference from that of HISUI, which can be used to investigate the cross-calibration uncertainty caused by hour-scale differences in overpass time. On 2022/08/03 and 2023/06/23, Terra/MODIS provided two daytime scenes, both of which were applied to the cross-calibration to increase the number of independent reference data sources.

Most of the MODIS data had a substantial zenith angle far away from nadir, except for Aqua/MODIS on 2021/08/02 and 2022/02/19. This very likely requires BRDF correction when comparing these data with HISUI nadir observations. Section III-D explains the treatment of the geometry difference.

III. METHOD

A. Overall Algorithm

The overall workflow, visualized in Fig. 2, is a combination of a cross-calibration of the corresponding bands between

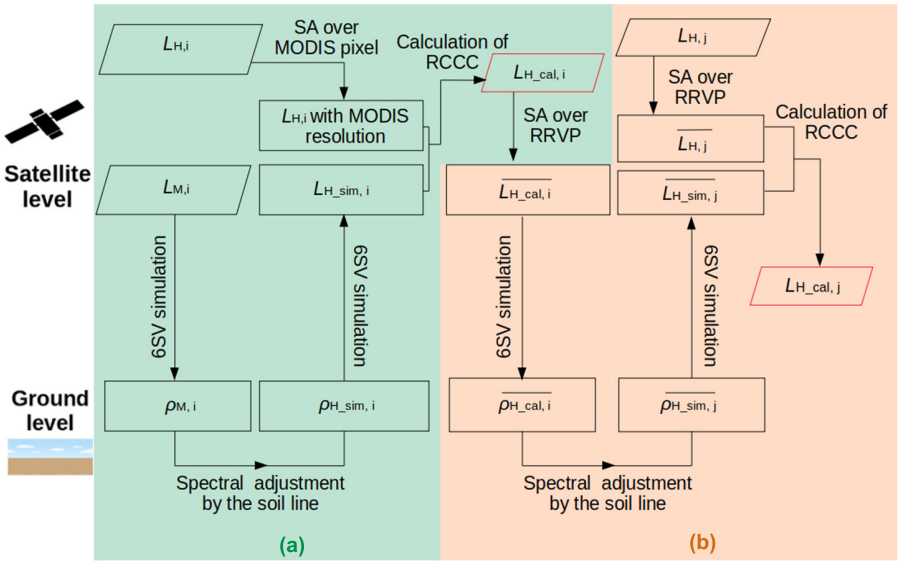

Fig. 2. Flowchart of the algorithm. (a) Cross-calibration part obtains RCCCs and calibrates the TOA radiance $L_{\mathrm{H_cal},i}$ of HISUI solely for corresponding MODIS bands i $(= 1, 2, \ldots, 7)$. (b) Subsequent inter-band calibration part obtains RCCCs for all HISUI bands $L_{\mathrm{H_cal},j}$ $(j = 1, 2, \ldots, 185)$.

TABLE II
BAND CORRESPONDENCE BY THE APPROXIMATE CENTER WAVELENGTH BETWEEN HISUI AND MODIS (ON BOTH TERRA AND AQUA)

MODIS band number	MODIS center wavelength (nm)	HISUI band number	HISUI center wavelength (nm)
1	645	25	645
2	859	46	855
3	469	7	465
4	555	16	555
5	1240	87	1251
6	1640	118	1638
7	2130	157	2125

MODIS and HISUI (Table II), and an inter-band calibration of the other HISUI bands by referring to the calibrated HISUI bands in the cross-calibration.

The cross-calibration phase [Fig. 2(a)] includes two-way radiative transfer simulation [transformation between the top of atmosphere (TOA) and bottom of atmosphere (BOA)], spectral band adjustment that corrects differences in the SRF between MODIS and HISUI, and spatial aggregation (SA) of HISUI pixels over a MODIS pixel.

First, the band i MODIS TOA radiance $L_{\mathrm{M},i}$ acquired at the point closest to the center of our study site was converted to BOA surface reflectance $\rho_{\mathrm{M},i}$ using radiative transfer simulation (6SV 2.1; see Section III-B for details).

Second, a linear regression between surface reflectance in corresponding MODIS and HISUI bands (i.e., the soil line [21], [22]) at RRVP was used to simulate HISUI-like surface reflectance $\rho_{\mathrm{H_sim},i}$

$$\rho_{\mathrm{H_sim},i} = \alpha_i \times \rho_{\mathrm{M},i} + \beta_i \tag{1}$$

where α_i and β_i are the slope and intercept of the regression line at band i, respectively. The regression line was created in advance by convolution of the in situ surface spectra (obtained from the past field surveys) with the MODIS and HISUI SRFs. The MODIS SRF was provided by the MODIS Characterization Support Team [33], and the HISUI SRF was estimated by a Gaussian function with a given center wavelength and full-width at half-maximum (FWHM) values. The determination of the center wavelength and FWHM of HISUI is explained in Section III-C.

After the spectral band adjustment, backward 6SV simulation converted $\rho_{\mathrm{H_sim},i}$ to the simulated HISUI-like TOA radiance $L_{\mathrm{H_sim},i}$. In contrast, the original HISUI TOA radiance values $L_{\mathrm{H},i}$ over the MODIS pixel of interest were aggregated with the weighting function of the MODIS IFOV [23]. The RCCC was defined by the ratio of the aggregated $L_{\mathrm{H},i}$ to $L_{\mathrm{H_sim},i}$

$$\mathrm{RCCC}_i = \frac{L_{\mathrm{H},i}}{L_{\mathrm{H_sim},i}} \tag{2a}$$

which is the RCCC derived from MODIS band i $(= 1, 2, \ldots, 7)$ and is used to calibrate the original $L_{\mathrm{H},i}$. The calibrated radiance $L_{\mathrm{H_cal},i}$ is obtained by

$$L_{\mathrm{H_cal},i} = \frac{L_{\mathrm{H},i}}{\mathrm{RCCC}_i}. \tag{2b}$$

The following inter-band calibration phase [Fig. 2(b)] repeats similar two-way radiative transfer simulation and spectral band adjustment by the regression line, but between the calibrated HISUI bands by (2b) and uncalibrated HISUI bands.

To mitigate the influence of spatiotemporal heterogeneity and residual cross-track sensor RNU even after the previous calibration activity [27], HISUI values within an area measuring 1 × 1 km around the RRVP center were averaged (i.e., the SA over RRVP). The SA value of the calibrated TOA radiance $\overline{L_{\mathrm{H_cal},i}}$ was converted to the BOA surface reflectance $\overline{\rho_{\mathrm{H_cal},i}}$ using 6SV, which was further used to simulate surface reflectance for the uncalibrated band

$$\overline{\rho_{\mathrm{H_sim},j}} = \alpha_{ij} \times \overline{\rho_{\mathrm{H_cal},i}} + \beta_{ij} \tag{3}$$

TABLE III
ATMOSPHERIC PARAMETERS FOR ATMOSPHERIC CORRECTION ON EACH SCENE DATE

Parameter	2021/08/02	2021/10/01	2022/02/19	2022/08/03	2023/06/23
Ozone (atm-cm)	0.2822	0.2899	0.3079	0.3054	0.3313
AOD at 550 nm	0.2395	0.0554	0.0371	0.1629	0.1211
Angstrom parameter	1.713	1.777	1.790	1.713	1.642
Water vapor (g/cm^2)	1.620	0.480	0.185	2.355	1.335

TABLE IV
WAVELENGTH SHIFT δ_λ AND FWHM OF HISUI DATA FOR EACH SCENE DATE DEPENDING ON THE CROSS-TRACK PIXEL LOCATION x_s

Parameter	2021/08/02	2021/10/01	2022/02/19	2022/08/03	2023/06/23
VNIR x_s	243	783	252	611	632
SWIR x_s	237	764	246	597	617
VNIR δ_λ	-4.1 nm	-3.2 nm	-3.2 nm	-3.2 nm	-3.2 nm
SWIR δ_λ	+2.8 nm	+4.0 nm	+4.5 nm	+4.6 nm	+4.6 nm
VNIR FWHM	17.4 nm	16.9 nm	16.5 nm	16.2 nm	16.2 nm
SWIR FWHM	23.7 nm	25.5 nm	24.3 nm	24.3 nm	24.3 nm

where α_{ij} and β_{ij} are the slope and intercept, respectively, of the regression line from reference band i to uncalibrated band j. Backward 6SV simulation converted $\overline{\rho_{\mathrm{H_sim},j}}$ to the TOA radiance $\overline{L_{\mathrm{H_sim},j}}$, which was compared with the original SA TOA radiance $\overline{L_{\mathrm{H},j}}$ to obtain the RCCC

$$\mathrm{RCCC}_{ij} \equiv \frac{\overline{L_{\mathrm{H},j}}}{\overline{L_{\mathrm{H_{sim}},j}}}. \tag{4}$$

Note that there are multiple reference band candidates ($i = 1, 2, \ldots, 7$) to derive an RCCC. Given that the coefficient of determination for the linear regression is generally higher when the selected wavebands are closer to each other [17], we relied more on the result from the closer reference waveband by the following weighted average:

$$\mathrm{RCCC}_j = \sum_i \frac{\mathrm{RCCC}_{ij}}{|\lambda_i - \lambda_j|} \Bigg/ \sum_i \frac{1}{|\lambda_i - \lambda_j|}. \tag{5}$$

Then, we obtained the calibrated TOA radiance $L_{\mathrm{H_cal},j}$ for all HISUI bands

$$L_{\mathrm{H_cal},j} = \frac{L_{\mathrm{H},j}}{\mathrm{RCCC}_j}. \tag{6}$$

B. Radiative Transfer Simulation

Our cross- and inter-band calibration approach includes BOA surface reflectance estimation (i.e., atmospheric correction) from the TOA radiance and vice versa. The 6SV 2.1 radiative transfer code was used for this purpose, with sensor geometry (Table I) and atmospheric parameters collected from the other data sources.

For ease of operation, atmospheric parameters from the re-analysis data freely provided by Copernicus Atmosphere Monitoring Service of the European Centre for Medium-Range Weather Forecast (ECMWF) [34], [35] and the U.S. National Centers for Environmental Prediction (NCEP) and the U.S. National Center for Atmospheric Research (NCAR) [36] were automatically collected for each day via the Google Earth Engine [37], at the time closest to the HISUI overpass. Required parameters were total column ozone concentration, total aerosol optical depth (AOD) at 550 nm, angstrom parameters, and column water vapor concentration; the first three were obtained or retrieved from the ECMWF, and the last was obtained from NCEP/NCAR. These are summarized in Table III. Further detailed settings for radiative transfer simulations are found in [17].

C. HISUI Spectral Response Function

Using a spectral analysis of atmospheric absorption bands, Yamamoto et al. [6], [7] reported a center wavelength shift and fluctuation in the FWHM for HISUI bands depending on the observation period. Until 2022/01/04, the wavelength shift and FWHM fluctuation features also showed cross-track dependency (the so-called "spectral smile"). Although the wavelength shift is within one band of spectral resolution, it is not negligible for our analysis, especially around narrow atmospheric absorption bands. Accurate estimation of the center wavelength and FWHM is also essential for generation of the regression lines required in (1) and (3).

Therefore, we precisely checked the wavelength shift δ_λ and FWHM fluctuations by referring to [6] and [7]. We confirmed the cross-track pixel location x_s of the study area on the entire HISUI swath (VNIR: 1024 pixels; SWIR: 1000 pixels) using the pre-georeferenced images (Level 1R). At the determined cross-track location, the typical values of δ_λ and FWHM for each period were obtained, as shown in Table IV. Because three different results were available for the SWIR domain corresponding to different O_2 and CO_2 absorption bands [6], [7], the average values were used.

The corrected center wavelength $\lambda_{0,\mathrm{cal}}$ for each band can be estimated by

$$\lambda_{0,\mathrm{cal}} = \lambda_0 - \delta_\lambda \tag{7}$$

where λ_0 is the nominal value of the center wavelength. The HISUI SRF was then given by the following Gaussian

function:

$$\mathrm{SRF}(\lambda) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(\lambda-\lambda_{0,\mathrm{cal}})^2}{2\sigma^2}\right\} \tag{8}$$

$$\sigma = \frac{\mathrm{FWHM}}{2\sqrt{2\ln 2}}. \tag{9}$$

D. BRDF Correction

Differences in observation angle between HISUI and MODIS and changes in the solar incidence angle during their observation interval change the apparent surface reflectance and should result in a cross-calibration bias. Our previous study [17] avoided this bias by selecting only quasi-SNO pairs from the data-rich archives of Hyperion and MODIS. However, such an approach was difficult to adopt in this study due to the limited amount of HISUI data available so far. Although HISUI usually provides an approximately nadir observation, the matched MODIS observation is often available only at the edge of a swath, making the zenith angle of the sensor view substantial (Table I).

This issue can be corrected by a BRDF, as known for calibration of instruments such as MODIS, the Visible Infrared Imaging Radiometer Suite, and very-high-resolution sensors [38], [39], [40]. To investigate the performance of BRDF correction, we compared three cross-calibration approaches: no BRDF correction (i.e., Lambertian assumption), BRDF correction using the MODIS BRDF product, and BRDF correction based on the field observation results for RRVP reported in [41].

The BRDF correction was achieved by substituting the MODIS surface reflectance at MODIS geometry $\rho_{\mathrm{M},i}(\theta_{\mathrm{S,M}}, \theta_{\mathrm{V,M}}, \varphi_{\mathrm{M}})$ in (1) by that at HISUI geometry $\rho_{\mathrm{M},i}(\theta_{\mathrm{S,H}}, \theta_{\mathrm{V,H}}, \varphi_{\mathrm{H}})$, based on the c-factor approach [42]

$$\rho_{\mathrm{M},i}\left(\theta_{\mathrm{S,H}}, \theta_{\mathrm{V,H}}, \varphi_{\mathrm{H}}\right) = c \times \rho_{\mathrm{M},i}\left(\theta_{\mathrm{S,M}}, \theta_{\mathrm{V,M}}, \varphi_{\mathrm{M}}\right) \tag{10}$$

$$c = \frac{\hat{\rho}\left(\theta_{\mathrm{S,H}}, \theta_{\mathrm{V,H}}, \varphi_{\mathrm{H}}\right)}{\hat{\rho}\left(\theta_{\mathrm{S,M}}, \theta_{\mathrm{V,M}}, \varphi_{\mathrm{M}}\right)} \tag{11}$$

where $\theta_{\mathrm{S,M}}$ and $\theta_{\mathrm{S,H}}$ are the solar zenith angles at the MODIS and HISUI observation times, $\theta_{\mathrm{V,M}}$ and $\theta_{\mathrm{V,H}}$ are the sensor zenith angles of MODIS and HISUI, and φ_{M} and φ_{H} are the azimuth angles of solar rays relative to MODIS and HISUI, respectively. $\hat{\rho}$ denotes the BRDF at the designated geometry, which is represented in the MODIS BRDF model [32], [43] as

$$\hat{\rho}(\theta_{\mathrm{S}}, \theta_{\mathrm{V}}, \varphi) = f_{\mathrm{iso}} + f_{\mathrm{vol}} K_{\mathrm{RT}} + f_{\mathrm{geo}} K_{\mathrm{LS}} \tag{12}$$

$$K_{\mathrm{RT}} = \frac{\left[\left(\frac{\pi}{2} - g\right)\cos g + \sin g\right]}{\left[\cos\theta_{\mathrm{S}} + \cos\theta_{\mathrm{V}}\right]} - \frac{\pi}{4} \tag{13}$$

$$\cos g = \cos\theta_{\mathrm{S}}\cos\theta_{\mathrm{V}} + \sin\theta_{\mathrm{S}}\sin\theta_{\mathrm{V}}\cos\varphi \tag{14}$$

$$K_{\mathrm{LS}} = O - \sec\theta_{\mathrm{S}} - \sec\theta_{\mathrm{V}} + \frac{(1+\cos g)\sec\theta_{\mathrm{S}}\sec\theta_{\mathrm{V}}}{2} \tag{15}$$

$$O = \frac{1}{\pi}(t - \sin t\cos t)(\sec\theta_S + \sec\theta_V) \tag{16}$$

$$\cos t = 2\frac{\sqrt{G^2 + (\tan\theta_S\tan\theta_V\sin\varphi)^2}}{\sec\theta_S + \sec\theta_V} \tag{17}$$

$$G = \sqrt{\tan^2\theta_S + \tan^2\theta_V - 2\tan\theta_S\tan\theta_V\cos\varphi} \tag{18}$$

where θ_{S} is the solar zenith angle, θ_{V} is the sensor zenith angle, and φ is the azimuth angle of solar rays relative to a sensor. The parameters f_{iso}, f_{vol}, and f_{geo} were provided by the 16-day composite MODIS daily product (MCD43A1). We averaged 16 adjacent data points that include the day of interest and obtained the parameters as Table V. Note that the 2021/10/01 parameters were the average of six data points due to the data unavailability in the MCD43A1 daily product for ten days.

In (12), $\hat{\rho}$ can be represented in the modified Rahman–Pinty–Verstraete (RPV) model used in [41] as

$$\hat{\rho} = \rho_0(\cos\theta_S\cos\theta_V(\cos\theta_S + \cos\theta_V))^{k-1} h F(g) \tag{19}$$

$$h = 1 + \frac{1-\rho_0}{1+G} \tag{20}$$

where ρ_0 and k are the model parameters and g is the phase angle calculated by (14). In the original RPV model, $F(g)$ is the Henyey–Greenstein phase function factor

$$F(g) = \frac{1-\Theta^2}{\left(1+2\Theta\cos g + \Theta^2\right)^{\frac{3}{2}}} \tag{21}$$

but is replaced in the modified algorithm [44] by

$$F(g) = \exp(-b_M\cos g). \tag{22}$$

The parameter sets (ρ_0, k, Θ) or $(\rho_0, k, b_{\mathrm{M}})$ are needed to simulate the bidirectional reflectance factor. In this case, we used $\rho_0 = 0.179$, $k = 0.800$, and $b_{\mathrm{M}} = -0.254$, as reported in [41], obtained at the RRVP point spatiotemporally closest to our study site. Bruegge et al. [41] also reported that the parameters are spectrally invariant unless there has been no recent rainfall, and temporally consistent within ~2% uncertainty. Therefore, we used the parameters for all bands and scenes commonly.

In addition to the c-factor approach, 6SV was also run in the "BRDF mode" that includes the MODIS BRDF and RPV models. Because the phase function of the RPV model in 6SV was implemented with (21), we estimated Θ from b_{M} using a secant method (*optimize.newton* method of the *scipy* library in Python 3.8.10), by solving the following equation:

$$f(\Theta) = \exp(-b_M\cos g) - \frac{1-\Theta^2}{\left(1+2\Theta\cos g + \Theta^2\right)^{\frac{3}{2}}} = 0. \tag{23}$$

In three cases of Terra MODIS on 2022/02/19, 2022/08/03 (19:10), and 2023/06/23 (18:55), $f(\Theta)$ had a positive local minimum, and no solution was obtained even after a large number of iterations (i.e., a convergence error). In such cases, we decided to use a Θ value that gives the local minimum as closest to the solution, by solving the first derivative of $f(\Theta)$ set to zero.

IV. Results

A. Cross-Calibration Between Corresponding Bands

As explained above, RCCCs for seven HISUI bands corresponding to MODIS bands were first derived for each

TABLE V
MODIS BRDF PARAMETERS ON EACH SCENE DATE

Band	2021/08/02	2021/10/01	2022/02/19	2022/08/03	2023/06/23
1 f_{iso}	0.421	0.487	0.391	0.405	0.378
1 f_{vol}	0.155	0.157	0.120	0.262	0.262
1 f_{geo}	0.057	0.105	0.061	0.010	0.051
2 f_{iso}	0.461	0.519	0.430	0.440	0.413
2 f_{vol}	0.205	0.197	0.160	0.263	0.284
2 f_{geo}	0.050	0.095	0.048	0.004	0.045
3 f_{iso}	0.227	0.264	0.201	0.214	0.196
3 f_{vol}	0.074	0.078	0.059	0.143	0.134
3 f_{geo}	0.031	0.062	0.033	0.004	0.028
4 f_{iso}	0.367	0.429	0.339	0.353	0.328
4 f_{vol}	0.128	0.132	0.092	0.240	0.226
4 f_{geo}	0.055	0.100	0.060	0.011	0.049
5 f_{iso}	0.450	0.490	0.426	0.422	0.402
5 f_{vol}	0.189	0.199	0.152	0.201	0.234
5 f_{geo}	0.050	0.080	0.033	0.009	0.040
6 f_{iso}	0.491	0.529	0.483	0.445	0.439
6 f_{vol}	0.245	0.246	0.186	0.204	0.271
6 f_{geo}	0.047	0.074	0.027	0.000	0.038
7 f_{iso}	0.437	0.457	0.375	0.384	0.380
7 f_{vol}	0.191	0.216	0.162	0.177	0.213
7 f_{geo}	0.044	0.056	0.009	0.004	0.026

observation date (Table VI). If the HISUI data were already well calibrated, the RCCCs were expected to fluctuate just around unity (i.e., RCCC = 1.000 means perfect consistency with MODIS), which was found on 2021/08/02, 2021/10/01, and 2022/02/19 with BRDF correction, ranging approximately from 0.95 to 1.05.

However, a substantial difference from unity was often found. The first difference was an error caused by the absence of BRDF correction. The RCCCs on 2021/10/01 without BRDF correction abruptly dropped from the previous date (2021/08/02) and reappeared on the next date (2022/02/19).

Given that such an implausible pattern in time series was not found in BRDF-corrected RCCCs, the apparent change in RCCCs on 2021/10/01 without BRDF correction was likely an error caused by the large zenith angle in the MODIS data (Table I). As such, the RCCCs without BRDF correction were not reliable when used with an off-nadir counterpart and were not used for the determination of final RCCCs in this study.

The second aspect was further problematic and observed on 2022/08/03 and 2023/06/23. The BRDF-corrected result showed that VNIR (from M1 to M4) RCCCs decreased to 0.7–0.8, whereas SWIR (from M5 to M7) RCCCs increased to 1.2–1.3, making an obvious discrepancy between the VNIR and SWIR spectral domains. This is highly likely related to the ISS maneuver events on 2022/03/20 and 2022/06/25 that were also partly reported in [6], and are further discussed in Section V-A. At this stage, RCCCs before and after the maneuver events were separately treated because HISUI radiometric sensitivity changed greatly after spring 2022.

Fig. 3 visualizes the time-series change of RCCCs (Table VI) for each HISUI band corresponding to MODIS. This also confirmed that BRDF-uncorrected RCCCs (i.e., solid lines) unreasonably varied from day to day. Before the maneuver events, if we assume BRDF-corrected lines are closer to reality, the absolute differences of BRDF-uncorrected RCCCs from the lines were 0.1–0.2 (10%–20%), which were rough estimations of the cross-calibration uncertainty without BRDF correction. BRDF correction (dashed or dashed-dotted lines) generally created consistent RCCCs regardless of the reference sensor (Terra or Aqua) and BRDF model (MODIS or RPV) within ±0.05 (∼5%) before the maneuvers. After the maneuvers, earlier-overpassing Terra/MODIS ("MOD A" represented by the dark blue lines) tended to show patterns dissimilar from those of the other results. This is probably due to the larger difference in overpass time from HISUI making atmospheric conditions more different, resulting in higher uncertainty in the cross-calibration. This confirmed that more coincident overpassing counterparts are preferable, if the archive contains sufficient data.

Note that such discrepancies did not necessarily occur when the early-overpassing (17:00–18:00 in UTC; 10:00–11:00 in local summer time) Terra/MODIS data were used. For example, RCCCs derived from early-overpassing Terra/MODIS data on 2021/10/01 seemed to be consistent with those from Aqua/MODIS. The outcome is expected to depend on seasonal and meteorological conditions (e.g., atmospheric stability). Because this stage lacked deterministic criteria to distinguish RCCCs affected greatly by the atmospheric condition difference, and a larger sample size was needed, RCCCs derived from the early-overpassing reference were also included for determination of the final RCCCs.

Fig. 4 shows mean RCCCs from BRDF-corrected results for each date, with standard errors shown by light shading. These results also confirmed an obvious sensitivity alteration of HISUI data before and after the ISS maneuvers. The pre-maneuver data show a slight degradation in SWIR sensitivity from 2021 to 2022. The degradation tendency as time passed exceeded the 1σ standard error level. However, when taking the other uncertainty sources inherently arising from our cross-calibration approach into consideration, there seems to

TABLE VI
RCCCs FOR SEVEN REFERENCE BANDS OF HISUI FOR EACH SCENE DATE

Band	MYD Lambertian	MOD Lambertian	MYD MBRDF	MOD MBRDF	MYD RBRDF	MOD RBRDF
			2021/08/02			
M1	0.963	1.112	0.981	0.991	0.994	1.021
M2	0.966	1.109	0.992	1.006	1.003	1.005
M3	0.930	0.980	0.934	0.905	0.947	0.936
M4	0.979	1.092	0.992	0.974	1.005	1.018
M5	1.003	1.126	1.034	1.012	1.046	1.012
M6	-	1.076	-	0.977	-	0.961
M7	0.909	1.041	0.941	0.940	0.953	0.932
			2021/10/01			
M1	0.797	0.829	1.005	1.037	0.980	1.022
M2	0.785	0.799	0.991	1.007	0.974	0.993
M3	0.787	0.820	0.933	0.967	0.915	0.956
M4	0.807	0.837	1.002	1.031	0.973	1.011
M5	0.817	0.787	1.027	0.990	1.018	0.981
M6	-	0.784	-	0.992	-	0.977
M7	0.773	0.760	0.962	0.953	0.966	0.948
			2022/02/19			
M1	1.022	1.071	0.988	0.984	0.996	0.997
M2	0.995	1.017	0.967	0.959	0.965	0.935
M3	0.946	0.936	0.928	0.893	0.937	0.906
M4	1.026	1.065	0.993	0.977	1.006	1.007
M5	1.015	1.000	0.992	0.964	0.981	0.914
M6	-	0.974	-	0.950	-	0.888
M7	0.932	0.905	0.918	0.904	0.901	0.825
			2022/08/03			
M1	0.820	(A) 0.786, (B) 0.887	0.800	(A) 0.952, (B) 0.884	0.771	(A) 0.887, (B) 0.834
M2	0.834	(A) 0.779, (B) 0.884	0.817	(A) 0.940, (B) 0.887	0.782	(A) 0.886, (B) 0.816
M3	0.746	(A) 0.764, (B) 0.806	0.739	(A) 0.882, (B) 0.822	0.712	(A) 0.831, (B) 0.793
M4	0.825	(A) 0.810, (B) 0.886	0.807	(A) 0.972, (B) 0.887	0.780	(A) 0.902, (B) 0.849
M5	1.395	(A) 1.285, (B) 1.493	1.361	(A) 1.498, (B) 1.466	1.306	(A) 1.464, (B) 1.363
M6	-	(A) 1.263, (B) 1.372	-	(A) 1.467, (B) 1.378	-	(A) 1.441, (B) 1.239
M7	1.284	(A) 1.285, (B) 1.325	1.258	(A) 1.497, (B) 1.314	1.202	(A) 1.475, (B) 1.204
			2023/06/23			
M1	0.980	(A) 0.755, (B) 1.017	0.773	(A) 0.838, (B) 0.792	0.797	(A) 0.758, (B) 0.824
M2	0.974	(A) 0.714, (B) 0.985	0.788	(A) 0.812, (B) 0.789	0.783	(A) 0.721, (B) 0.781
M3	0.844	(A) 0.744, (B) 0.876	0.710	(A) 0.784, (B) 0.741	0.729	(A) 0.734, (B) 0.772
M4	0.948	(A) 0.768, (B) 0.995	0.753	(A) 0.832, (B) 0.782	0.788	(A) 0.766, (B) 0.832
M5	1.592	(A) 1.173, (B) 1.659	1.316	(A) 1.309, (B) 1.352	1.275	(A) 1.189, (B) 1.302
M6	-	(A) 1.159, (B) 1.520	-	(A) 1.316, (B) 1.257	-	(A) 1.176, (B) 1.186
M7	1.473	(A) 1.126, (B) 1.483	1.269	(A) 1.273, (B) 1.273	1.176	(A) 1.144, (B) 1.155

Notes: RCCCs were defined by the ratio of the aggregated original HISUI radiance to HISUI-like radiance simulated from MODIS for six cases: Aqua (MYD) and Terra (MOD), without BRDF correction (i.e., Lambertian assumption), with BRDF correction by the MODIS model (MBRDF columns) or the modified Rahman et al. model (RBRDF columns). From the top row for each date, RCCCs at HISUI bands 25, 46, 7, 16, 87, 118, and 157 that correspond to MODIS bands 1 to 7 (M1 to M7) are summarized. (A) and (B) on 2022/08/03 and 2023/06/23 denote the different MODIS/Terra observations, respectively.

be no strong evidence for significant degradation of the SWIR sensitivity (further uncertainty discussion is provided in Section V-B).

The discrepancy between 2022/08/03 and 2023/06/23 after the maneuvers was more prominent, and had an approximate 10%–15% difference in RCCCs. Validation of the two RCCCs for these dates is provided in Section V-A.

B. Inter-Band Calibrations for All HISUI Bands

The obtained RCCCs for the HISUI bands corresponding to MODIS were applied to the original HISUI data to calibrate the seven bands, which were used as references for the other band calibrations described in this section.

Each date in Fig. 5 (top) compares RCCCs derived from different reference sensors (Terra or Aqua) and different BRDF models (no BRDF, MODIS BRDF, and RPV model). Each date in Fig. 5 (bottom) compares the surface reflectance derived from the HISUI data calibrated with RCCCs and the original HISUI data. Except for 2021/08/02, RadCalNet surface reflectance data were also available (green lines).

As discussed in Section IV-A, RCCCs and derived surface reflectance without BRDF correction (solid lines) often showed results inconsistent with those obtained with

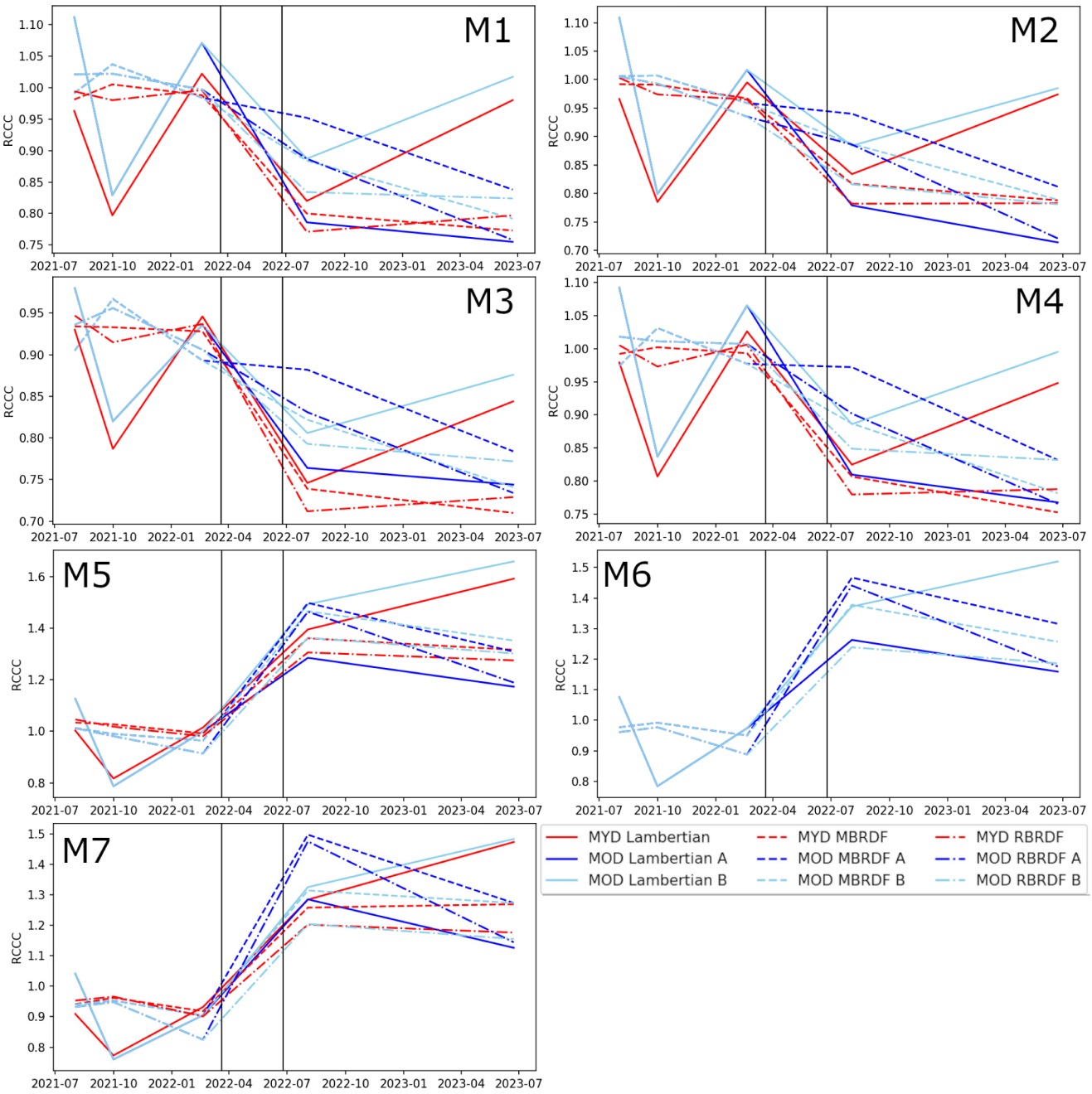

Fig. 3. Time series of RCCCs for HISUI bands corresponding to MODIS. M1: HISUI band 25 (645 nm); M2: HISUI band 46 (859 nm); M3: HISUI band 7 (469 nm); M4: HISUI band 16 (555 nm); M5: HISUI band 87 (1240 nm); M6: HISUI band 118 (1640 nm); and M7: HISUI band 157 (2130 nm). Solid lines represent no BRDF consideration (Lambertian assumption), dashed lines represent BRDF correction with the MODIS product, and dashed–dotted lines represent BRDF correction by the method in [38]. Red lines are results from Aqua/MODIS (MYD), and blue lines are results from Terra/MODIS (MOD—if two observations are available, the earlier observation is dark blue, and the later observation is sky blue). The vertical black lines represent the ISS maneuver events on 2022/03/20 and 2022/06/25.

BRDF correction and RadCalNet data. Typical examples are Terra on 2021/08/02, Terra and Aqua on 2021/10/01, and Terra and Aqua on 2023/06/23, in which solid lines (the surface reflectance without BRDF correction) do not match the lines with BRDF correction and RadCalNet, but rather match the apparent MODIS reflectance without BRDF correction (blue or red circles). This also supports the importance of BRDF correction when off-nadir counterpart data are used.

Except for strong atmospheric absorption bands shown by the gray areas, RCCCs typically fluctuated within the range of ± 0.05–0.1 along the wavelength. The fluctuating RCCCs were essential for calibrating the inter-band variability of the original HISUI data shown by black lines into smooth surface reflectance spectra that are consistent with the RadCalNet spectra. The waveband-weighted average (5) contributed to creation of the smoothed spectra. In our previous approach [17], only one or two adjacent reference

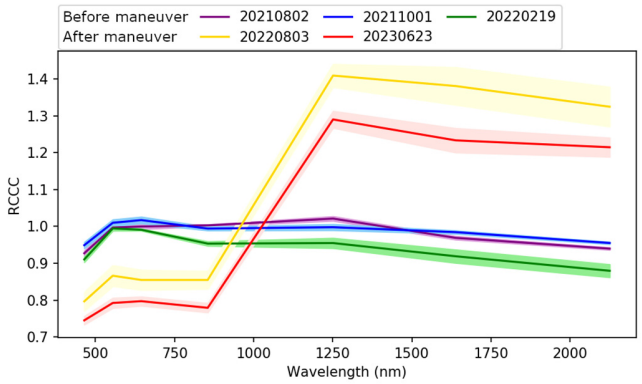

Fig. 4. Spectral patterns of RCCCs (solid lines) for HISUI bands corresponding to MODIS (469, 555, 645, 859, 1240, 1640, and 2130 nm) for each scene date (before the ISS maneuvers: 2021/08/02, 2021/10/01, and 2022/02/19; after the ISS maneuvers: 2022/08/03 and 2023/06/23). Light shading represents standard errors of the RCCC estimation from different reference sensors (Terra and Aqua) and different BRDF models (MODIS product or [41]).

bands were used for inter-band calibration and this sometimes resulted in gaps in the spectral pattern at the wavelengths where the reference bands were switched. The new strategy (i.e., waveband-weighted average) eliminated such gaps, which is a minor but important improvement in our algorithm.

The atmospheric absorption bands (gray areas) were determined by 6SV simulation on the 2023/06/23 Aqua overpass time as the wavelengths where transmittance was less than 0.7. In such wavebands, strong absorption occurs mainly by water vapor, whose spatiotemporal variation decreases the accuracy of atmospheric correction and may increase uncertainty in the derived RCCCs. In fact, RCCC spectral patterns particularly around 1400–1500 and 1800–2000 nm largely varied from day to day, which seemed to include errors caused by differences in atmospheric conditions between observations.

The large discrepancy between the VNIR and SWIR domains after the maneuver events was also confirmed in RCCCs for all HISUI bands (2022/08/03 and 2023/06/23). Similar to the results described in Section IV-A, VNIR RCCCs were around 0.7–0.8, whereas those in the SWIR domain were 1.3–1.5. As a result, the original surface reflectance (and TOA radiance, which is not shown) around boundary wavebands between VNIR and SWIR (900–975 nm) was greatly disconnected. Our inter-band calibration corrected that to a smoothed surface reflectance pattern. Note that a subtle inconsistency between VNIR and SWIR was also observed in the original surface reflectance before the maneuvers, but only at the edge of SWIR (band 59, 901 nm). This is probably because sensitivity at the edge of observable bands is poor and may also be poor in the opposite edge at longer wavebands (around 2300–2500 nm), both of which can be corrected by the RCCCs.

Fig. 6 visualizes average RCCCs with BRDF correction for each date, with standard errors shown by light shading. Before the maneuvers (top), consistent RCCC spectra were generally observed within standard errors, suggesting systematic (i.e., temporally less variant) inter-band variation in the HISUI radiometric sensitivity. Exceptions were found at the edge of observable bands around 400–500, 900, and 2100–2500 nm, and near the strong atmospheric absorption bands around 920–970, 1100–1200, 1350–1500, 1700–2000, and 2400–2500 nm shown by the gray areas. Although not shown by the gray areas, 720 and 820 nm also have weak water vapor absorption, and thus weak inconsistency was observed. Less consistency among RCCCs at the edge of observable bands can be attributed to a low signal-to-noise ratio in those spectral regions, which is a generally known tendency for hyperspectral sensors (e.g., [45]).

More different spectra of RCCCs were observed between 2022/08/03 and 2023/06/23 (bottom). For VNIR, RCCCs on 2022/08/03 were systematically larger than those on 2023/06/23 in almost all bands, which can be attributed to a similar bias in the cross-calibration results (Fig. 4). SWIR RCCC patterns between the two dates looked less consistent, not only in the edges of observable bands and atmospheric absorption bands but also in most bands (see the validation in Section V-A).

V. Discussion

A. Determination of RCCCs

This study provided the first results of cross- and inter-band calibration of HISUI, deriving RCCCs under various conditions over five days. Inter-band variation in the original HISUI radiance (and surface reflectance) data and the discontinuity between the VNIR and SWIR domains were substantially improved by calibration using the RCCCs (Fig. 5), which will improve the applicability of HISUI to spectral analysis to search for narrow-band absorption or reflection features. The calibrated surface reflectance was also consistent with the other validation data source (i.e., RadCalNet), which supports the overall validity of our cross- and inter-band calibration approach.

This study also revealed the problematic discrepancy between the VNIR and SWIR domains after the ISS maneuver events (2022/03/20 and 2022/06/25). Yamamoto et al. [6] reported that the spectral characteristics of HISUI can be classified into five major durations (P1: before 2021/05/23; P2: 2021/05/27–2021/09/23; P3: 2021/09/30–2022/01/04; P4: 2022/01/05–2022/03/29; and P5: after 2022/04/01). Among them, P5 has been characterized by "the appearance of prominent vertical stripes" after the ISS maneuvers, which was a noticeable sensitivity variation of HISUI along cross-track directions, particularly around x_s ranging from 240 to 420 nm. The cross-track dependency (vertical stripes) has been already corrected by HISUI calibrations [27]; however, no report has addressed the influence of the event on spectral dependency. This research is also a prompt report on the HISUI sensitivity alteration after the maneuver events in the context of the spectral aspect. The RCCCs obtained on 2022/08/03 and 2023/06/23 will be given to the HISUI calibration working team and will soon contribute to radiometric correction after the maneuver events. To date, no official reports have described the mechanism of this phenomenon; however, we guess that the maneuver events might have unintentionally

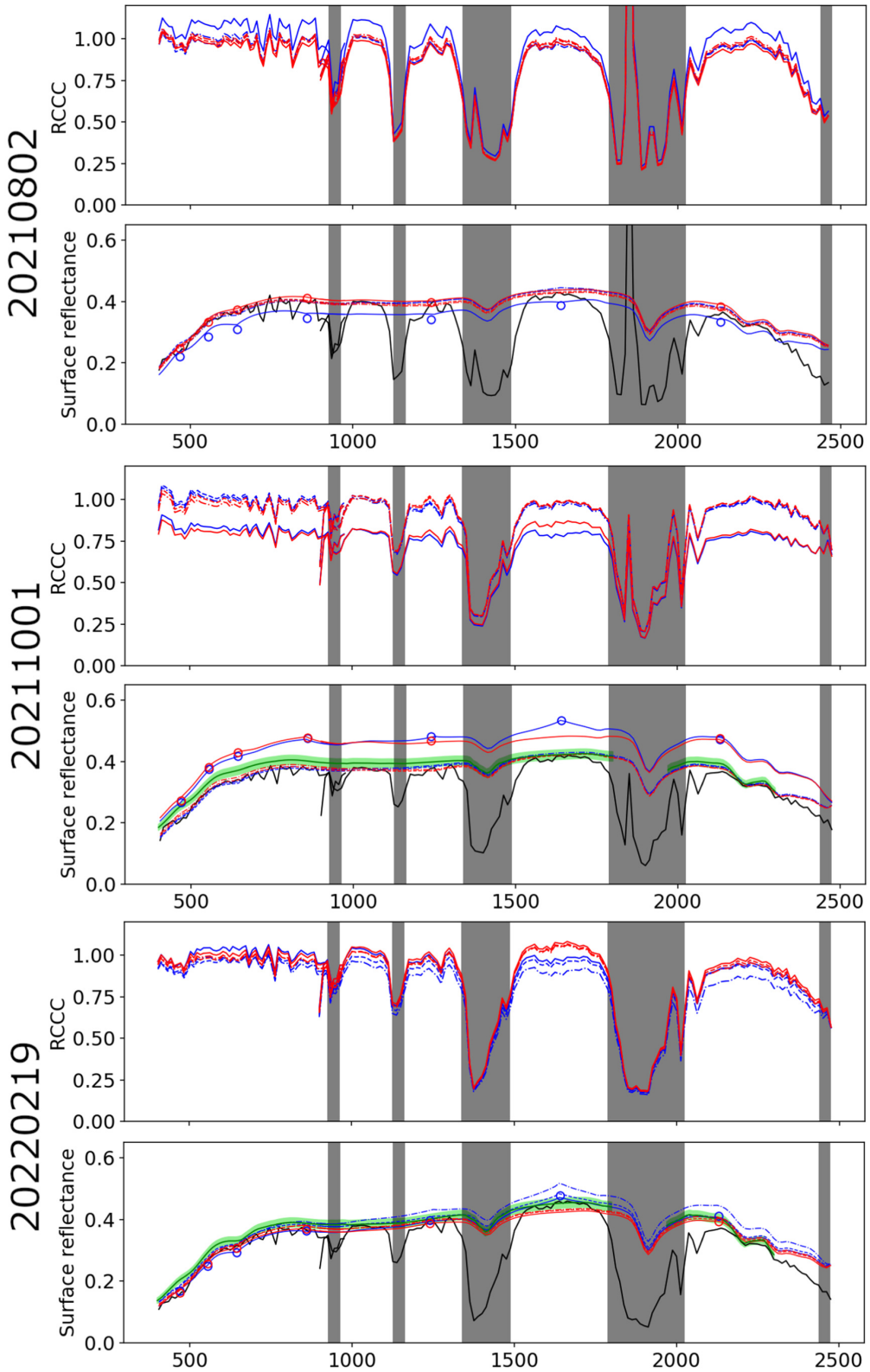

Fig. 5. RCCCs and surface reflectance for all the HISUI bands. For each date, (top) shows RCCCs, and (bottom) shows surface reflectance. Line styles are the same as in Fig. 3, but the RadCalNet surface reflectance (green solid lines with uncertainty area shown by light green shading) and the surface reflectance derived from HISUI original radiance (black solid lines) were added to bottom. The gray rectangular areas represent atmospheric absorption bands mainly due to water vapor, where uncertainty in the derivation of RCCCs and surface reflectance is expected to be high.

allowed stray solar light to enter the optical path and possibly cause non-fatal but significant changes in some parts of HISUI's optical system.

Other than the abrupt HISUI sensitivity change after the maneuver events, a smaller-scale sensitivity alteration of HISUI along time-series data before the maneuver events

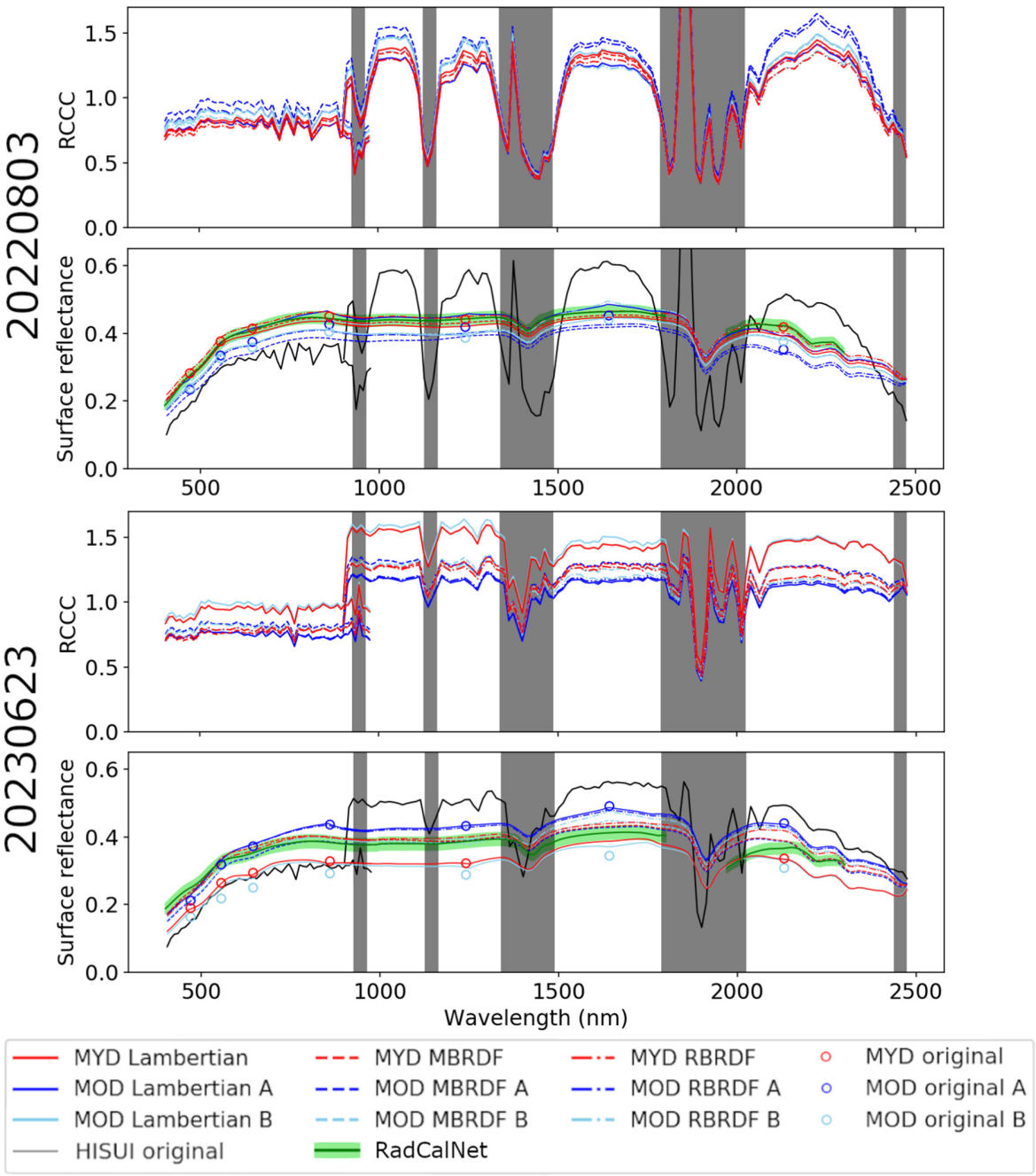

Fig. 5. *(Continued.)* RCCCs and surface reflectance for all the HISUI bands. For each date, (top) shows RCCCs, and (bottom) shows surface reflectance. Line styles are the same as in Fig. 3, but the RadCalNet surface reflectance (green solid lines with uncertainty area shown by light green shading) and the surface reflectance derived from HISUI original radiance (black solid lines) were added to bottom. The gray rectangular areas represent atmospheric absorption bands mainly due to water vapor, where uncertainty in the derivation of RCCCs and surface reflectance is expected to be high.

(Figs. 4 and 6) was also indicated, but the difference seemed to be within the uncertainty range of the applied algorithm.

A larger difference in the overall RCCC patterns after the maneuvers between 2022/08/03 and 2023/06/23 motivated us to make additional validation, using other RadCalNet sites (Gobabeb, Baotou, and Baotou Sand). Five additional HISUI datasets acquired after the maneuvers were used to derive RCCCs through a procedure similar to that for RRVP (details are described in the Appendix), although they did not cover the full wavelength range of HISUI due to the limited spectral range of RadCalNet data for those calibration sites.

Comparison of the additional RCCCs with those derived from RRVP (Fig. 7) showed that all the RCCC patterns from the additional RadCalNet sites were more consistent with that of 2023/06/23 than 2022/08/03, implying that the former RCCCs were more reliable. A potential reason for the higher uncertainty on 2022/08/03 may be a systematic error caused by unexpected disturbances in the atmosphere and/or ground surface, such as very thin cirrus clouds that cannot be visually interpreted, a local-scale water vapor anomaly, and abrupt changes in soil moisture due to intermittent rain.

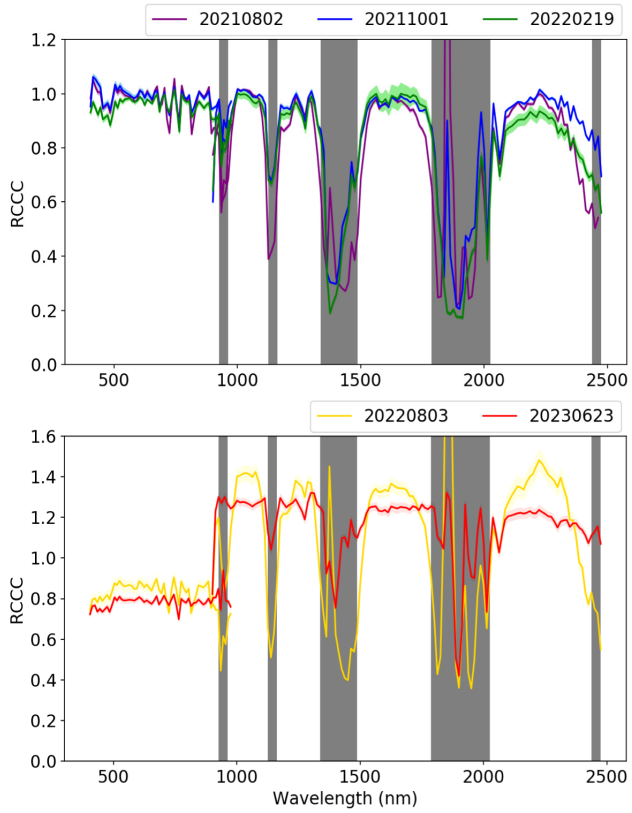

Fig. 6. Average RCCCs for each date. (Top) RCCCs before the maneuver events and (bottom) RCCCs after the maneuver events. Line styles are the same as in Fig. 4, and the gray areas show the atmospheric absorption bands.

TABLE VII
PROPOSED RCCCS FOR HISUI BANDS CORRESPONDING TO MODIS BANDS (M1–M7) BEFORE AND AFTER THE ISS MANEUVERS

MODIS band	Before maneuvers	After maneuvers
M1	1.003	0.797
M2	0.983	0.779
M3	0.929	0.745
M4	1.000	0.792
M5	0.991	1.291
M6	0.958	1.234
M7	0.925	1.215

TABLE VIII
UNCERTAINTY ESTIMATED BY RELATIVE STANDARD ERRORS (%) OF THE DETERMINED HISUI RCCCS ARISING FROM DIFFERENT BRDF MODELS AND COUNTERPART SENSORS FOR EACH MODIS BAND

Scene date	M1	M2	M3	M4	M5	M6	M7
2021/08/02	0.69	0.22	0.78	0.82	0.60	0.48	0.34
2021/10/01	0.87	0.50	0.89	0.87	0.81	0.44	0.32
2022/02/19	0.26	0.62	0.82	0.60	1.43	1.95	1.98
2022/08/03	3.15	2.83	3.21	3.27	2.21	3.69	4.05
2023/06/23	1.54	1.60	1.52	1.71	1.75	2.66	2.11
Before maneuvers[a]	0.38	0.27	0.48	0.45	0.57	0.66	0.65
After maneuvers[b]	1.54	1.60	1.52	1.71	1.75	2.66	2.11

[a]Uncertainties before the maneuvers were estimated based on the uncertainty propagation from each date.
[b]Uncertainties after the maneuvers was represented by those on 2023/06/23 because RCCCs on 2023/06/23 are recommended for use for the period after the maneuvers.

RCCC results from both 2022/04/02 and 2023/02/01 at the Gobabeb site gave RCCC patterns similar to those on 2023/06/23 at RRVP. This implies that the multiple maneuvers (2022/03/20 and 2022/06/25) did not create multiple alterations in RCCCs, and RCCCs on 2023/06/23 are assumed to be common for the after-maneuver duration (i.e., 2022/03/20 to present).

Note that the RCCC patterns on 2023/06/23 seem to have a cut-off level (upper limit) around 1.3. Given the increased SWIR sensitivity (1.3–1.5) after the maneuver events and the high solar zenith angle in summer, the cut-off pattern may be related to the saturation of the sensor sensitivity. We have preliminarily confirmed that there were no saturation-flagged pixels that exceeded the sensor dynamic range over the study site. However, there were several pixels having nearly saturation-level digital numbers in surrounding pixels, suggesting that the high intensity of the desert scene after the maneuver events may be close to the upper limit of sensor sensitivity. The sensor sensitivity curve against the light intensity (i.e., sensor linearity) should be comprehensively checked using multiple calibration sites [15], [16] and/or a spatiotemporally homogeneous target having various brightness levels in the near future.

In any case, additional samples for cross- and inter-band calibrations are needed to confirm or model detailed RCCCs along time series. To determine the final RCCCs in this prompt report, one choice is to assign RCCCs obtained for each date with durations corresponding to the findings of [6] (P1–P2: 2021/08/02; P3: 2021/10/01; and P4: 2022/02/19). However, it is uncertain that the durations determined based on the characteristics of spectral smile [6] can be commonly applied to durations for radiometric characteristics. Another choice is to use the average of RCCCs from the three observations before the maneuver event, and to use RCCCs derived from 2023/06/23 based on the additional validation result (Table VII), which seems to be the best available estimation. Due to the limited amount of HISUI data available for appropriate calibration sites, we have not established a time-series model (i.e., sensor degradation function) for HISUI. Instead, the HISUI product will be updated in an ad hoc manner, by applying the proposed (or slightly refined) RCCCs (Table VII) to the current radiance data of HISUI based on (2b) in the near future.

B. Uncertainty

This work demonstrated that quasi-simultaneous off-nadir overpass counterparts can be used for cross-calibration with appropriate BRDF correction. The relaxed selection of the counterpart images can increase the opportunities for cross-calibration. The strategy will be useful for sensors that have a limited number of data archives of images acquired over calibration sites.

However, calibration uncertainty from calculations (i.e., BRDF correction) added to the original algorithm [17] is likely

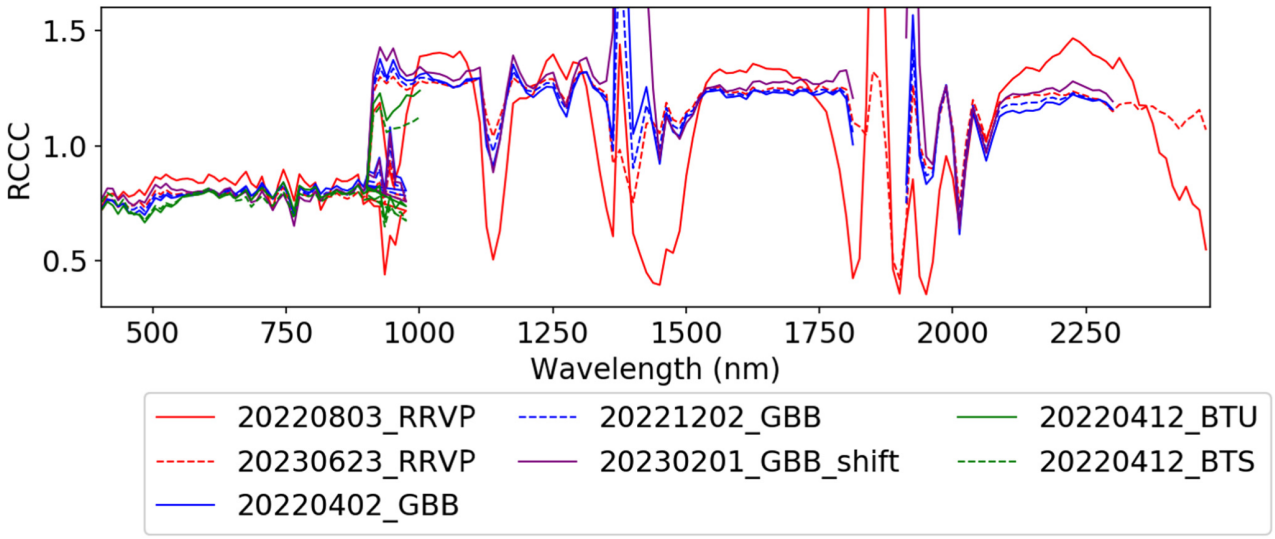

Fig. 7. Comparison of RCCCs derived from RRVP (red solid line: 2022/08/03; red dotted line: 2023/06/23) with those from the other RadCalNet sites (GBB: Gobabeb; BTU: Baotou; and BTS: Baotou Sand).

TABLE IX
ERROR BUDGET TABLE ESTIMATED FOR INTER-BAND CALIBRATION AT TYPICAL WAVEBANDS

Error	500 nm	700 nm	900 nm	1100 nm	1300 nm	1500 nm	1700 nm	2100 nm
HISUI reference bands (before maneuvers)	1.84	1.42	2.09	1.46	2.05	1.40	1.83	2.40
HISUI reference bands (after maneuvers)	2.06	1.64	2.34	1.65	2.31	1.82	2.53	2.95
Other error sources[a]	2.21	2.16	4.26	3.59	2.22	4.32	1.79	1.62
RSS (before maneuvers)	2.88	2.58	4.75	3.88	3.02	4.54	2.56	2.90
RSS (after maneuvers)	3.02	2.71	4.86	3.95	3.20	4.69	3.10	3.37

[a]Error sources other than HISUI reference bands (i.e., variability in atmospheric conditions, soil line influence, and solar irradiance) were adopted from [17] and used to estimate the total uncertainty as the root sum of squares (RSS).

to arise. Table VIII summarizes the relative standard error of the RCCCs averaged from different sensors (Terra or Aqua) and different BRDF models (MODIS BRDF or RPV) in the corresponding MODIS bands.

The standard errors were generally less than 1% on 2021/08/02, 2021/10/01, and 2022/02/19 (except for SWIR on 2022/02/19 ranging within 1%–2%). Larger standard errors were observed on 2022/08/03 and 2023/06/23, ranging from approximately 2%–4%. The worse standard errors can be mainly attributed to the inclusion of earlier Terra/MODIS data on the latter two dates, rather than the maneuver events. A substantially large standard error was observed on 2022/08/03 (with early-morning Terra/MODIS data), and in MODIS band 6, where Aqua/MODIS data were not available.

Related to the inclusion of early-morning Terra/MODIS data, the differences in the uncertainties between before (less than 1%) and after (approximately 2%–3%) the maneuvers provide a rough estimate of increased uncertainty when using relaxed condition counterparts. Inclusion of counterparts having intervals of more than 2 h and/or acquired in the early morning or late afternoon may result in an uncertainty increase of 1.4%–2.0% (calculated from the root difference square between before and after the maneuvers except for band 6), due to the different atmospheric conditions and largely different overpass times between sensors. Conversely, searching for the counterpart with a slightly strict condition (intervals less than 1.5 h except for Terra/MODIS on 2021/10/01) mitigates the uncertainty. The standard error before the maneuvers (0.27%–0.66%) can be assumed as the summation of uncertainties from reference sensor selection, slight differences in atmospheric conditions, BRDF model selection, and/or temporally fixed RPV parameters [41] for seasonally variable BRDF (within a range of ∼2%).

Mizuochi et al. [17] gave a comprehensive uncertainty estimation of the cross- and inter-band calibration approaches for quasi-SNO pairs, including uncertainty in MODIS reflectance, atmospheric conditions, soil lines, geolocation, solar irradiance, and reference band selection. They estimated the total uncertainty in the cross-calibration phase as 2.5%–3.2%, and that in the inter-band calibration phase as 2.5%–4.8%, except for strong absorption bands. Root sum squares (RSS) between the previous estimation in the cross-calibration phase (2.5%–3.2%) and newly introduced uncertainty (0.27%–0.66%, due to different reference sensors and off-nadir observation with relatively strict counterpart selection) are estimates of overall uncertainty in the

TABLE X
SPECIFICATIONS OF THE HISUI AND MODIS DATA OVER THE ADDITIONAL SITES ON EACH COLLECTION DATE (YYYY/MM/DD)

Parameter	2022/04/02	2022/04/12[a]	2022/04/12[a]	2022/12/02	2023/02/01
HISUI overpass time and solar zenith angle	08:24:52 (local time 10:24:52), 47.98°	02:27:12 (local time 10:27:12), 43.91°	02:27:12 (local time 10:27:12), 43.91°	07:49:15 (local time 09:49:15), 41.29°	07:40:40 (local time 09:40:40), 50.01°
HISUI scene cloud rate	0%	0%	0%	2%	0%
Terra/MODIS overpass time and solar zenith angle	08:20:00 (local time 10:20:00), 48.37°	03:40:00 (local time 11:40:00), 34.63°	03:40:00 (local time 11:40:00), 34.63°	08:45:00 (local time 10:45:00), 27.92°	09:00:00 (local time 11:00:00), 31.35°
Terra/MODIS sensor zenith angle	63.23°	24.75°	24.75°	22.10°	12.40°
Aqua/MODIS overpass time and solar zenith angle	-	-	-	13:05:00 (local time 15:05:00), 26.12°	-
Aqua/MODIS sensor zenith angle	-	-	-	26.12°	-
RadCalNet observation time	08:30:00 (local time 10:30:00)	02:30:00 (local time 10:30:00)	02:30:00 (local time 10:30:00)	08:00:00 (local time 10:00:00)	2023/02/07 08:00:00 (local time 10:00:00)
Target site	Gobabeb	Baotou	Baotou Sand	Gobabeb	Gobabeb[b]

[a]The same HISUI and MODIS scenes were used for Baotou and Baotou Sand because they are sufficiently close to each other.
[b]For this day, a strict overpass of HISUI over Gobabeb RadCalNet was not available, and the closest homogeneous area was used instead.

cross-calibration phase for the current study, which were 2.6%–3.2% (M1: 2.54%; M2: 3.23%; M3: 3.22%; M4: 2.62%; M5: 3.24%; M6: 2.66%; and M7: 2.83%). If the counterpart with the relaxed condition is used (represented by the "after maneuvers" column in Table VIII), the overall uncertainty in the cross-calibration phase can increase to 3.0–3.7% (M1: 2.95%; M2: 3.60%; M3: 3.54%; M4: 3.10%; M5: 3.64%; M6: 3.71%; and M7: 3.47%).

In the inter-band calibration phase, another minor update was the weighted average of all reference bands (5) to estimate RCCCs of other bands. Considering the uncertainty propagation from the derived cross-calibration uncertainty, uncertainty in the inter-band calibration phase caused by HISUI reference bands in typical wavelengths was estimated as given in Table IX. Combining the RSS with other error sources provided by [17] gives an overall uncertainty in the inter-band calibration of 2.6%–4.8%. Even if the relaxed counterpart was used, the uncertainty was typically less than 5% (2.7%–4.9%) except for the strong atmospheric absorption bands.

VI. CONCLUSION

This study provided the first cross- and inter-band calibration for the Japanese spaceborne hyperspectral sensor HISUI. An updated algorithm for cross- and inter-band calibration using wide-swath reference sensor (MODIS) data enabled us to obtain RCCCs to calibrate all HISUI bands. To fully use the limited number of HISUI observations over an appropriate calibration target (RRVP), quasi-simultaneous off-nadir reference data were included by applying the BRDF correction. This treatment mitigated the influence of different geometries and created robust RCCC patterns with less than 5% overall uncertainty except for the strong atmospheric absorption band. Using RCCCs substantially improved the inter-band variability in the original spectra of radiance or surface reflectance of HISUI data, making the spectra consistent with those of RadCalNet surface reflectance data. Investigation of time-series RCCCs further revealed the problematic discrepancy between the VNIR and SWIR spectral domains after the ISS maneuver events in March and June 2022. RCCCs after the maneuver events will be provided to the HISUI calibration working team and will soon contribute to correcting the reported sensitivity decrease in VNIR (RCCCs of 0.7–0.8) and increase in SWIR (RCCCs of 1.3–1.5).

APPENDIX

Solely to validate RCCCs after the maneuvers on 2022/08/03 and 2023/06/23, additional RCCC retrievals were conducted for five cases over three other RadCalNet sites: Gobabeb (15.11956E, 23.6002S), Baotou (109.6291E, 40.8514N), and Baotou Sand (109.6155E, 40.86587N). The HISUI and MODIS data we used are summarized in Table X.

Unlike RRVP, these other RadCalNet data cover only limited spectral regions (Gobabeb: 400–1820 and 1910–2310 nm; Baotou and Baotou Sand: 400–1000 nm). In addition, the Gobabeb HISUI scene acquired on 2023/02/01 did not strictly cover the RadCalNet point, and the neighboring northwestern area was used instead. Further, coincident RadCalNet data were not available, and data acquired on the nearest date (2023/02/07) were used to check the resulting HISUI surface reflectance.

The procedures to obtain RCCCs were similar to those for RRVP: coregistration with MODIS, determination of the SRF and band adjustment, radiative transfer simulations with atmospheric parameters obtained from ECMWF and NCEP/NCAR, BRDF correction using the MODIS MCD43A1 product, calculation of RCCCs for reference bands corresponding to MODIS and all HISUI bands, and comparison of surface reflectance patterns with RadCalNet data for double-checking.

In the process, band adjustment based on (1) and (3) requires in situ hyperspectral data, which were substituted for all the RadCalNet observations acquired on the target dates;

2022/04/02, 2022/12/02, and 2023/02/01 were used together for Gobabeb, and 2022/04/12 was used for Baotou and Baotou Sand. Correspondingly, the RCCC results relating to the lacking wavelengths in RadCalNet data (350–400, 1820–1910, and 2310–2500 nm for Gobabeb, and 350–400 and 1000–2500 nm for Baotou and Baotou Sand) were not reliable. As such, these results are not suited for the determination of final RCCCs for the full range of wavelengths, but they were still useful for validation of RRVP results retrieved on 2022/08/03 and 2023/06/23.

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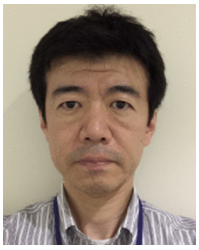

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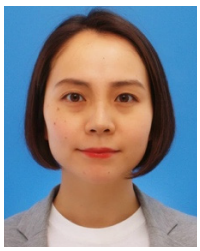

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